

# $\text{Li}_3\text{YCl}_6$ — crystalline transport benchmark

From experimental reference to a controlled three-model MD comparison

## 1 Literature

Asano et al. (2018)  
Experimental chloride SSE  
Room-temperature transport

## 2 Structure

Explicit Li/Y ordering  
 $2 \times 2 \times 2$  supercell  
240 atoms; three NNPs

## 3 MD protocol

400–1000 K; NVT  
1 fs; 50 ps equilibration  
500 ps production

## 4 Analysis

Li MSD and  $D(T)$   
Arrhenius  $E_a$   
Conditional  $\sigma_{NE}$

## 5 Conclusion

Thermal trend reproduced  
Absolute transport is  
strongly model dependent